

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Bhaskara Contest

(NMTC- at **JUNIOR LEVEL IX & X Standards**)

Saturday, 29th August 2009.

Note:

- 1) Fill in the response sheet with your Name, Class, the institution through which you appear in the specified places.
- 2) Diagrams are only visual aids; they are not drawn to scale.
- 3) You are free to do rough work on separate sheets.
- 4) Duration of the test: 2 p.m. to 4. p.m. - 2 hours.

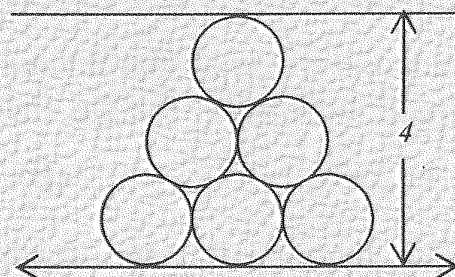
PART – A

Note:

- Only one of the choices A,B,C,D is correct for each question. Shade that alphabet of your choice in the response sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
- For each correct response you get 1 mark; **for each incorrect response you lose $\frac{1}{4}$ mark.**

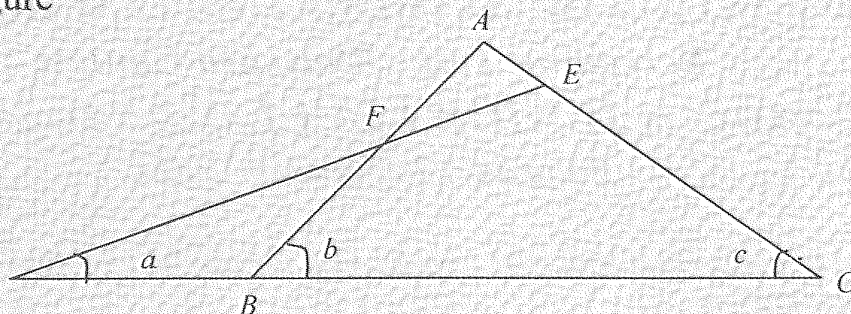
1. Six equal circles are arranged as in the figure. The height of the triangle is 4.

Then the radius of the circle is

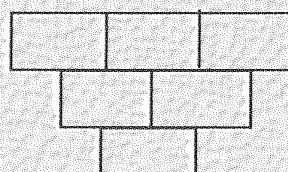
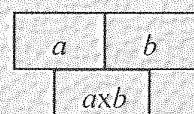


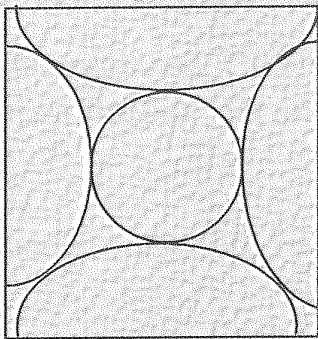
- (A) $(\sqrt{3}-1)$ (B) $\left(\frac{\sqrt{3}-1}{2}\right)$ (C) $\frac{4}{\sqrt{3}+1}$ (D) $\sqrt{3}$
2. The number of squares on a coordinate plane with one vertex at $A(-2, 2)$ and at least one of the coordinate axes as axis of symmetry of the square is
- (A) 3 (B) 5 (C) 6 (D) 7
3. The number of real solutions of the equation $(x+1)(3^x-2)=1$ is
- (A) 0 (B) 1 (C) 2 (D) more than 2

4. In the adjacent figure



- (A) $a + b + c = 180^\circ$ (B) $a + c = b$ (C) $2c + a = b$ (D) $2a + c = b$
5. A positive integer m is triangular if $m = 1 + 2 + 3 + \dots + n$ for some integer $n > 0$. If m is triangular, then $(8m + 1)$ is
- (A) also triangular (B) a perfect square
(C) a perfect cube (D) none of the above three
6. Let $f(n)$ be the number of non-congruent integer sided triangles with perimeter n ($f(3) = 1, f(4) = 0, f(5) = 1$). Then $f(10)$ is
- (A) 2 (B) 3 (C) 4 (D) greater than 4
7. Ananya wrote all the integers of 1 to 6 digits. She uses only the two digits 0 and 5. In all, how many 0's did she use? (leading zeros not allowed in the numbers)
- (A) 256 (B) 128 (C) 129 (D) 200
8. The diagonal BD of a quadrilateral $ABCD$ bisects the angle $\angle ABC$. It is given that $AC = BC, \angle BDC = 80^\circ, \angle ACB = 20^\circ$. $\angle BAD$ is
- (A) 110° (B) 120° (C) 130° (D) 140°
9. The sum of the digits of a number ' n ' is 30. Which one of the following cannot be the sum of the digits of the number $n + 3$?
- (A) 6 (B) 15 (C) 21 (D) 24
10. A numerical puzzle is constructed with positive integers greater than 1 in each cell as shown in the figure. In the following grid with 6 cells which number cannot be filled in the bottom most cell



11. The number of values of a for which $x^2 + 3ax + 2009 = 0$ has two integer roots is
 (A) 3 (B) 4 (C) 6 (D) 8
12. If two circles of radii 13 and 15 intersect and the length of the common chord is 24, then the distance between their centres is
 (A) 10 (B) 12 (C) 14 (D) 16
13. Four students Anu, Banu, Chandu, Deeraj make the following statements.
 Anu : Banu, Chandu and Deeraj are Math Majors.
 Banu : Anu, Chandu and Deeraj are commerce majors
 Chandu : Both Anu and Banu are lying
 Deeraj : Anu, banu and Chandu are telling the truth.
 Then the number of students telling the truth is
 (A) 0 (B) 1 (C) 2 (D) 3
14. Given a triangle of area 80cm^2 , circles of radii 2cm are drawn with centres at the vertices of the triangle. What is the area of the triangle outside of these circles (in cm^2)
 (A) $\pi \rightarrow 70$ (B) $\pi \rightarrow 76$ (C) $78 - \pi$ (D) none of these
15.  In the diagram, four semi circles of radius 1cm each with their centres at the mid points of the sides of square are drawn. A circle is inscribed touching all the four semi-circles. The radius of the inner circle is
 (A) $\sqrt{2} - 1$ (B) $\frac{\pi}{2} - 1$ (C) $\sqrt{3} - 1$ (D) $2\sqrt{2} - 2$

PART – B

Note:

- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{2}$ mark.

16. The sum of 100 digit numbers having the sum of their digits equal to 3 is
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18. Each term of sequence is the sum of its preceding two terms from the third term onwards. The second term of the sequence is -1 and the 10^{th} term is 29 . The first term is _____
19. The number of positive integers satisfying the inequality $2000 < \sqrt{n(n+2)} < 2009$ is _____
20. The number of pairs of positive integers (x,y) satisfying the equation $x^2 + y^2 + 2xy - 2008x - 2008y - 2009 = 0$ is _____
21. Given $a, b, c \geq 0$ and $a + b + c = 1$ then the maximum value of $\sqrt[3]{a+b} + \sqrt[3]{b+c} + \sqrt[3]{c+a}$ is _____
22. All the numbers from 1 to 2009 are written one after the other. The number of digits in this number is _____
23. A square is described on the hypotenuse of a right angled triangle away from the right angled vertex. The ratio of the perpendiculars drawn from the intersection point of the diagonals of this square to the legs of the right angled triangle is _____
24. PMT is a tangent to the circle APC at the point P ; $CNAT$ is a diameter, to which PN is drawn perpendicular and AM is perpendicular to PT . the ratio of AM to AN is _____
25. If the length of a diagonal of a cube is $\sqrt{12}$ cm; then the area of each of its faces is _____
26. The sum of the digits of the number $(1000^{20} - 20)$ expressed in decimal notation is _____
27. A point is taken on the hypotenuse of a right triangle equidistant (x) from the legs. The point divides the hypotenuse into parts 30cm and 40cm. Then x is equal to _____ cm
28. Two circles touch externally at C . AB is a direct common tangent touching the circles at A, B . If $AC = 14$ cm, $BC = 48$ cm then the length of the direct common tangent is _____
29. In a quadrilateral there consecutive sides are of lengths 2, 3 and 4. A circle of radius 1.2 is inscribed in the quadrilateral. The area of the quadrilateral is _____
30. a, b, c, d and e are positive reals. If $a - 2c + e = 0$, $b - 2c + d = 0$, $c - 2d + e = 0$ then the set consisting of the largest and the smallest numbers among a, b, c, d, e is _____